

Business Process Technologies and Management

Exercises

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Assignment 2

1. Export BPMN 2.0 XML from Signavio
2. Write a small program that converts your XML to a Woflan TPN (see examples at <https://www.win.tue.nl/woflan/>)
 - The program must run on `lehre.bpm.in.tum.de`
 - Commandline: `./convert your.bpmn > your.tpn`
 - Warning: if the commandline does not exactly like that, no points will be awarded
 - Explanation of the commandline:
 - your program has to be named "convert";
 - convert is an executable file, it may contain python, ruby, nodejs, ...
 - convert has 1 parameter, a valid bpmn file
 - convert outputs tpn to STDOUT (onto the terminal)
 - `> your.tpn` redirects STDOUT to a file
3. Install Woflan, check if it loads: <https://www.win.tue.nl/woflan/>
 - Conduct a reachability analysis
 - Create a PDF documenting the analysis with screenshots
4. Use the parsing knowledge you gained when parsing the BPMN XML to create your own commandline or web-tool that conducts a reachability analysis
 - Output the results
 - Commandline: `./reachability your.tpn > your.txt`
 - Warning: if the commandline does not exactly like that, no points will be awarded
 - Explanation of the commandline:
 - your program has to be named "reachability"
 - reachability is an executable file, it may contain python, ruby, nodejs, ...
 - reachability has 1 parameter, a valid tpn file
 - reachability outputs the reachability to STDOUT (onto the terminal)
 - `> your.txt` redirects STDOUT to a file
 - The reachability output has to be in a SPECIFIC output format, or no points are awarded (see next slide)

Assignment 2 - Output Format

Format Specification:

Tx ... [LABEL]

Ty ... [LABEL]

M₁₁₁₁Px₁Py₁Transitions

Mx₁₁₁[NUM]₁₁[NUM]₁₁[Tx]→[Mx]₁₁[Ty]→[My]

Interpretation:

Mx: final | deadlock

...

Format Example:

T1 ... Collect Registrations

T2 ... Send Invitations

M₁₁₁₁P1₁P2₁Transitions

M0₁₁₁1₁₁0₁₁T1→M1

M1₁₁₁0₁₁1₁₁

M1: final

Assignment 2 - Bonus

Bonus on top of fully functional and correctly formatted reachability analysis (5 bonus points):

```
T1 ... Collect Registrations
```

```
T2 ... Send Invitations
```

```
M_ _ _ _ P1_ P2_ Transitions
```

```
M0_ _ _ 1_ _ 0_ _ T1->M1
```

```
M1_ _ _ 0_ _ 1_ _
```

```
M1: final
```

Bounded: 1-Bounded

Safe: true

Live: false

Quasi-live: false

Livelock: Mx->My (or empty)

Soundness: all places had tokens, all transitions fired at least once

Assignment 2 - Testing

- Test with various petri nets from the lecture slides (create tpn files for them)
 - Share tpn test files amongst each other for tests
- Test with the tpn that came from your own BPMN

Assignment 2 - TPN Format

```
place "P0" init 1;
place "P1";
...
trans "T1"~"Label" in "P0", "P2" out "P1";
...
trans "T1"~"Label"
    in "P0", "P2"
    out "P1";
```